Texas College of Management & IT

Bachelor of Computer Science (Hons) (Network Technology & Cybersecurity)

Assignment Title: **"** The Digital Identity Project **"**

Submitted By:Name: Animesh Babu Karki

LCID: LC00017003191

Year/Semester: BCS 2ND Year / 4TH Semester

Submitted To:

Lecturer Name: Dr. Ashish Gautam

Course: Web Desgining (WBD - 113)

Date of Submission: 25/07/2026

STUDENT DECLARATION

I declare that this mini project is my original work and completed by me.

Student Signature: Animesh Babu Karki

Date: 25/07/2026

FACULTY EVALUATION

Marks Obtained: _______

Comments:

Faculty Name: ______________________________

Signature: ______________________________

Date: ______________________________

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Introduction

This report explains how I planned, designed, and built my own personal portfolio website for the Digital Identity Project. The goal of this mini project was simple: build a website that shows off who I am, what I can do, and what I have made so far, while also proving that I understand the web development topics we covered in class. Those topics include semantic HTML, CSS layout tools like Flexbox and Grid, typography (the study of fonts and how they look), colour theory, making a site responsive (meaning it works on any screen size), and using a little bit of JavaScript to make things interactive.

The site was designed for an audience of prospective employers, collaborators, and recruiters within the fields of product Design, Networking, and Cybersecurity. Because of that, every decision I made while designing the site, from how the sections are ordered to which colours and fonts I picked, was made to help me look professional and trustworthy as fast as possible. Most people only spend a few seconds looking at a new webpage before deciding whether to keep reading, so I wanted those first few seconds to count.

Project Objectives

Before I started coding, I wrote down a short list of goals so I would know what I was actually trying to achieve. These goals came directly from the assignment brief, and I kept coming back to them throughout the project to check that I was still on track:

- Build a responsive, single-page-style personal portfolio that adapts fluidly across desktop, tablet, and mobile viewports.
- Demonstrate correct use of semantic HTML5 elements to support accessibility and search engine optimisation.
- Apply CSS Flexbox and Grid systems to construct a clean, consistent, and adaptable layout.
- Implement basic JavaScript interaction to enhance usability without compromising performance.
- Produce a professional, recruiter-ready personal website that showcases skills, activity, and projects, and provides clear contact information.

Core Understanding and Planning

Before writing a single line of code, I spent time planning out the structure of the website, the same way a professional developer would before starting a real project. This stage translated the abstract goal of "building a portfolio" into concrete decisions about structure, colour, and type.

The site follows a top-to-bottom narrative structure in the following order.

Header/Navigation → Hero → About → Activity → Highlighted Work → Side Projects → Ideas → Contact/Footer.

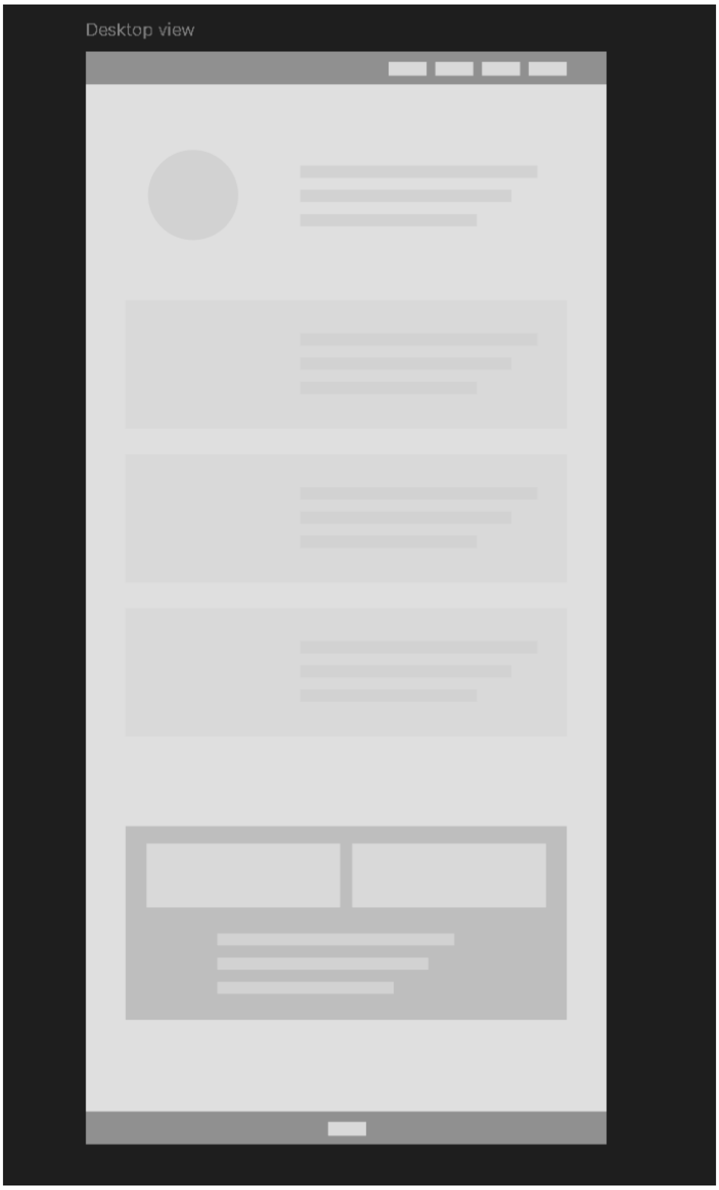

Figure 1. Desktop Wireframe

I sketched this out as a simple ‘Wireframe’ first, shown above, before touching any code. A wireframe is just a rough, low-detail sketch of where everything will go on the page. It let me test the ordering of the sections and move things around cheaply, without having to rewrite CSS every time I changed my mind.

This ordering was chosen deliberately to support the ‘F-Layout’ scanning pattern, the eye-tracking finding that most readers of Web content, including recruiters skimming a candidate's page, scan in two horizontal movements across the top of the content before dropping into a vertical scan down the left edge of the page.

To take advantage of this behaviour, the most important credibility signals were placed along the top horizontal band and the left-hand vertical axis of the page: My name and title sit in the upper-left of the header, the primary navigation runs horizontally across the top, and the hero section immediately beneath introduces my photo, role, focus areas, technical stack, experience, and location in a left-aligned list. This means a recruiter who reads only the top of the page and skims down the left margin will still absorb the candidate's name, discipline, and core qualifications before deciding whether to continue reading.

Colour Palette

The assignment brief recommended a maximum of three colours, split into a Primary, Secondary, and Accent role, and suggested a dark-mode aesthetic as one professional option. For this build, a light, high-contrast palette was chosen instead, on the reasoning that a bright background paired with a single saturated accent reads as clean and approachable for a design- and security-oriented audience, while still respecting the three-colour ceiling:

- Primary (background): White, #FFFFFF
- Secondary (text and structure): Dark grey/near-black body text, with supporting neutral greys of #F4F4F4, #EEEEEE, and #999999 used for section backgrounds, borders, and muted text.
- Accent: Blue, #2F6FED, used sparingly for links, highlighted keywords, and interactive states so that it draws the eye without overwhelming the page.

Keeping the palette this simple made the page feel calm instead of busy, and it meant the blue accent colour actually stood out whenever it appeared, because it was not competing with five other bright colours. If I had used more colours, the important clickable parts of the page could have easily gotten lost in the noise.

Typography

For the fonts, I ended up choosing *Tomorrow* (Google Fonts), a clean and modern sans-serif font, for all of my headings, because it gives the page a confident feel without shouting. For the body text, I used *Geist Mono* (Google Fonts), which is a monospace font (meaning every letter takes up the same amount of width, like on an old typewriter). I liked that this style hints at the technical and cybersecurity side of what I do, while still being very easy to scan through quickly.

Both fonts were self-hosted using the CSS @font-face rule rather than loaded from a third-party content delivery network, which removed an external network dependency and gave the page a small performance and privacy benefit.

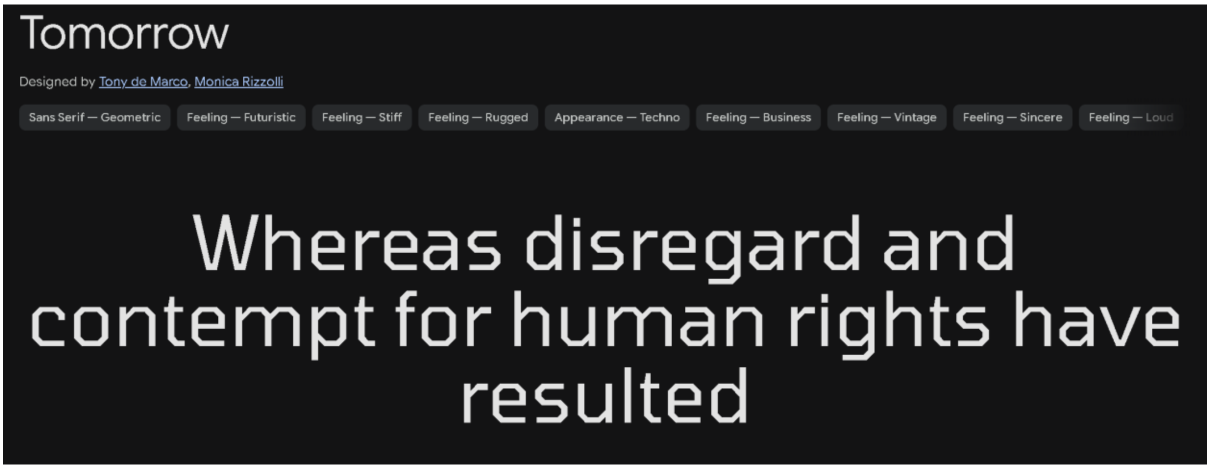

Figure 2: Font 'Tomorrow'

Figure 3. Font 'Geist Mono'

Development Steps and Code Structure

With the plan in place, development proceeded through four broad stages: semantic markup, layout and styling, interactivity, and responsiveness.

Semantic HTML Structure

Instead of wrapping everything in generic <div> tags, which do not tell a browser or a screen reader anything about what they contain, I used semantic HTML5 tags wherever they made sense, such as header, nav, main, section, and footer. Each big part of the page, the hero introduction, the About and Activity area, Highlighted Work, Side Projects, and Ideas, got its own section tag with a proper heading inside it. The navigation bar links directly down to each of these sections using anchor links, for example #about, #career, #work, #projects, and #ideas. Writing the HTML this way makes the page easier to use for people relying on screen readers, and it also helps search engines like Google understand what each part of my page is actually about.

```html
<section id="about">
  <div class="about-flex">
        <ul>
          <li><strong>Focus:</strong> <span class="fw-medium">Product Design</span> and <span class="fw-medium">Cybe
          <li><strong>Stack:</strong> Adobe Creative Suite (Ps, Ai, Ae, etc.), Figma, Python, and more.</li>
          <li><strong>Experience:</strong> 2+ years in Graphic Design, currently learning Networking & Cybersecurity
          <li><strong>Location:</strong> <span class="fw-medium">Kathmandu, Nepal</span></li>
        </ul>
        <p>
          <a href="https://github.com/ekka-np" target="_blank">GitHub</a> |
          <a href="https://www.instagram.com/animeshkarki_" target="_blank">Instagram</a> |
          <a href="mailto:animeshbabukarki@gmail.com">Email Me</a>
        </p>
      </div>
  </div>
</section>

<!-- Activity Section -->
<section id="career">
  <h2>Activity</h2>
  <p class="section-desc">Overview of my current academic and professional duties.</p>
  <ul class="career-list">
    <li>
      <span class="meta meta-date">November 2024 - Present</span>
      <span class="card-title">BCS</span>
      <span class="meta meta-org">Texas College of Management &amp; IT</span>
      <p>Currently pursuing a Bachelor's in Computer Science, specializing in Networking and Cybersecurity.</p>
      <p>I'm driven by the challenge of staying ahead of the ever-evolving cyber threats of tomorrow. My primary a
      <span class="meta meta-skills">Secure Architecture, SecOps, Threat Hunting</span>
    </li>
    <li>
      <span class="meta meta-date">October 2025 - Present</span>
```

Figure 4. HTML code

CSS Layout and Styling

CSS Flexbox was used to align and distribute items within one-dimensional groupings, such as the navigation bar, the social-link row, and the certification list, while CSS

Grid controlled the two-dimensional layouts needed for the hero section and the project galleries, where an image column and a text column needed to align together across a row. Throughout the whole site I kept the spacing between elements consistent, stuck to my small colour palette, and only used the one accent colour, so the page has a clear visual hierarchy (meaning it is obvious what to look at first) without needing heavy borders, boxes, or drop shadows everywhere.

JavaScript Functionality

Vanilla JavaScript was used sparingly and only where it added real usability value. A back-to-top control appears after the user scrolls past the hero section and smooth-scrolls the visitor back to the top of the page, reducing friction on a long, single-page layout. The second feature is a simple overlay on my project screenshots that makes it a bit harder for someone to right-click and save the original images, since some of that design and photography work is mine and I would rather people not casually save it.

Responsive Design Implementation

CSS Media queries were used throughout the stylesheet to reflow the grid-based sections into a single vertical column on narrow viewports, stack the navigation for touch input, and resize typography and imagery so that content remains legible without horizontal scrolling. The layout was checked at common breakpoints corresponding to typical phone, tablet, and desktop widths.

Testing and Quality Assurance

Once the site was built, I went through and tested it manually in both Chrome and Firefox to make sure it behaved the same way in each browser. I checked that every internal navigation link scrolled to the right section, that the page resized properly across different screen widths, that the external links to my GitHub, Instagram, and the pages of the organisations I work with all pointed to the right place, that the mailto link correctly opened an email addressed to me, and that every image and video on the page actually loaded instead of showing a broken link.

Challenges and Solutions

Not everything went smoothly on the first try. The three biggest headaches for me were getting my project screenshots to size correctly when they all had different width-to-height ratios, keeping the footer lined up properly across different screen sizes, and getting the spacing between sections to feel right, since early versions of the page felt either too cramped or had way too much empty space. None of these were solved in one attempt. I had to go back several times and adjust things like the sizing of my CSS Grid tracks, the object-fit property on my images (which controls how an image is cropped or scaled inside its box), and the amount of padding above and below each section, before the layout finally felt stable and consistent across the whole page.

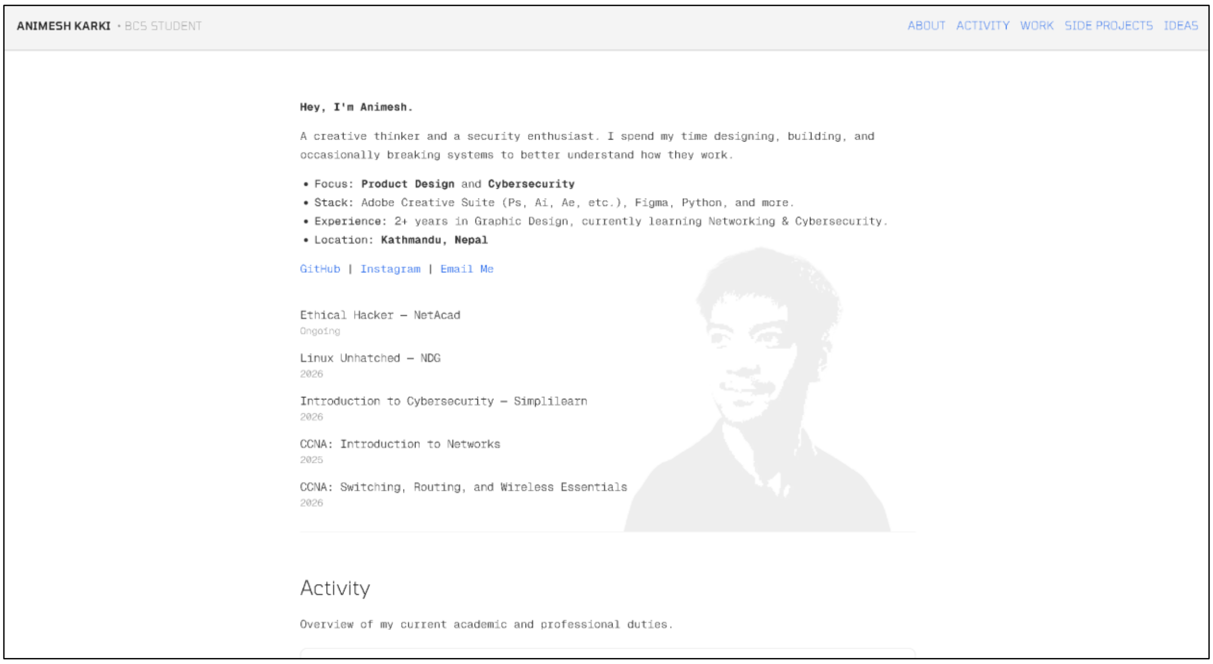

Figure 5. Final Website's Hero Section

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Future Improvements

The site is finished for this assignment, but there are a few things I would like to add or improve if I keep working on it afterwards:

- Adding a dedicated dark-mode toggle so visitors can switch between the current light theme and a dark aesthetic.

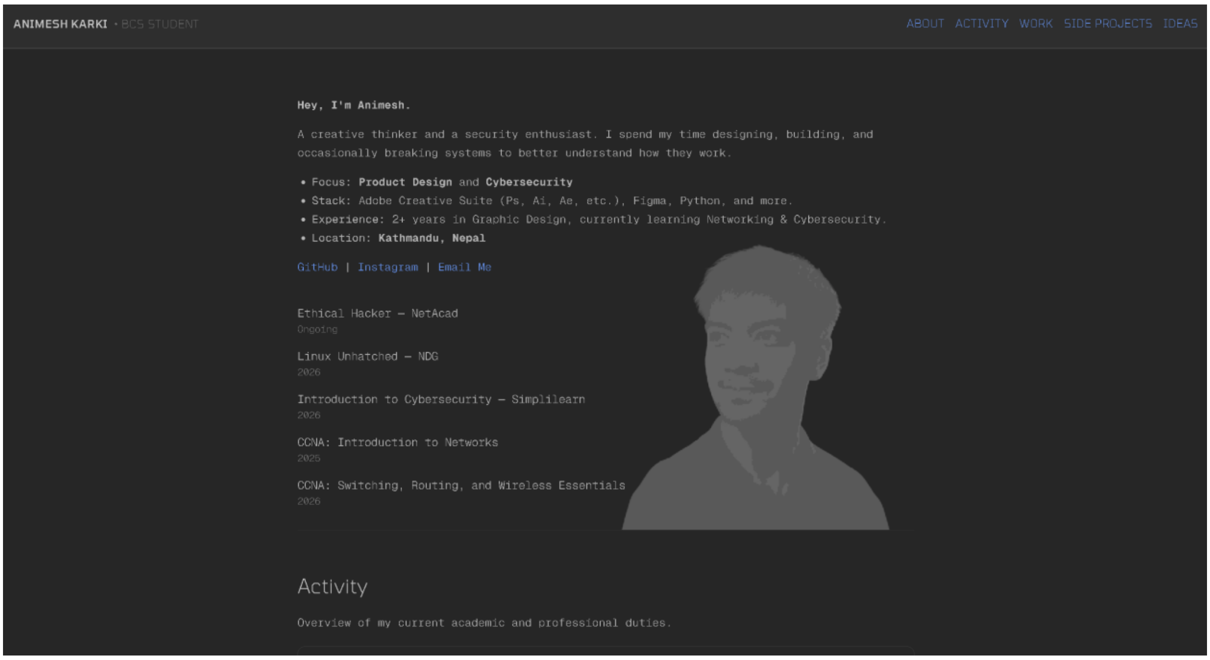

Figure 6. Idea for a 'Dark' mode for the Webpage

- Introducing a hamburger-style navigation menu for small screens to further simplify the mobile header.
- Improving accessibility through additional ARIA labels and a full keyboard-navigation audit.
- Adding subtle, performance-conscious animations to guide attention during scrolling.
- Doing more performance work, such as compressing images further and only loading images when they are about to appear on screen (called lazy loading), so my Lighthouse performance (Currently 97, See Appendix) score gets even closer to a perfect 100.

Submission Requirements

The following materials are provided in support of this submission, in accordance with the assignment's grading criteria.

GitHub Repository

The project's version-controlled source code was initialised as a Git repository and pushed to GitHub for public review at the following address:

https://github.com/ekka-np/web-dev

Live Deployment

The site is deployed and publicly accessible via GitHub Pages at the following address:

https://ekka-np.github.io/web-dev/

Screenshots

<u>Desktop View</u>

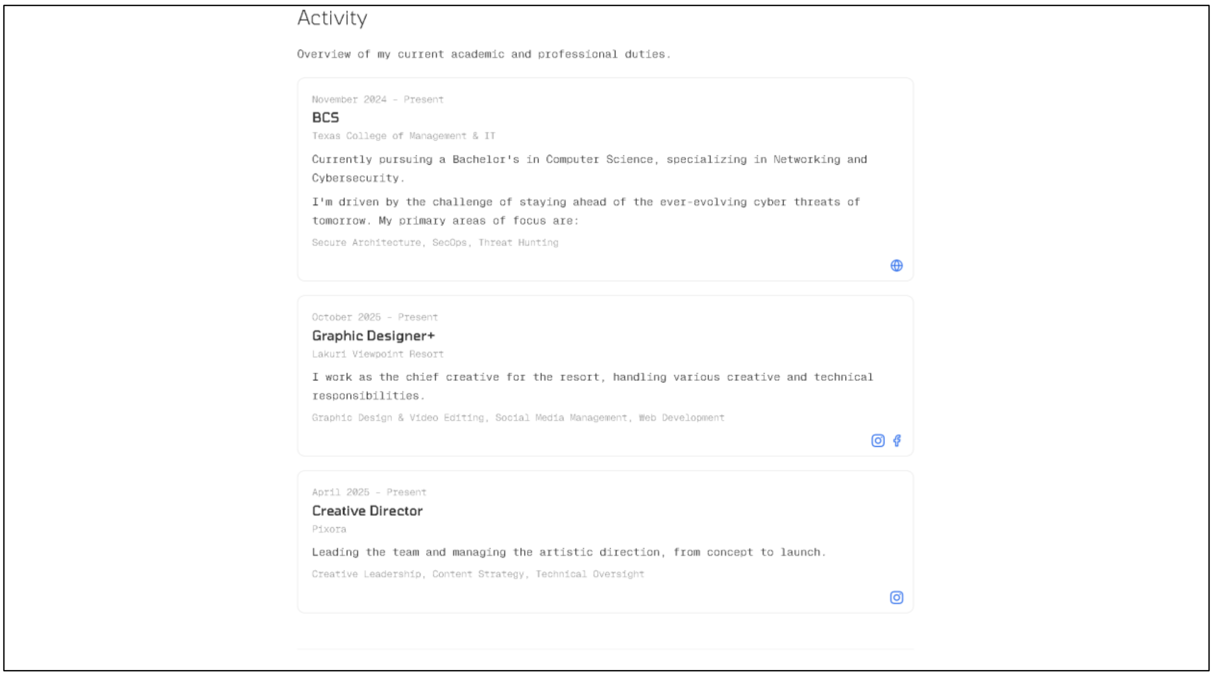

Figure 7. Desktop View (1)

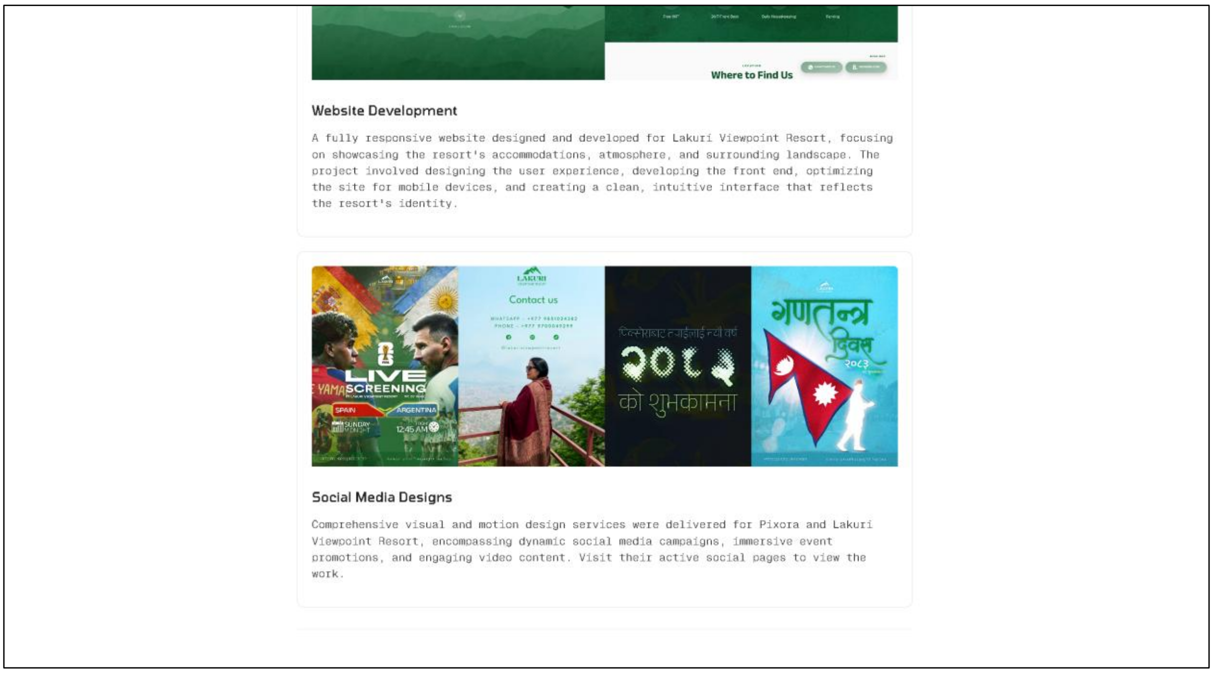

Figure 8. Desktop View (2)

Mobile View

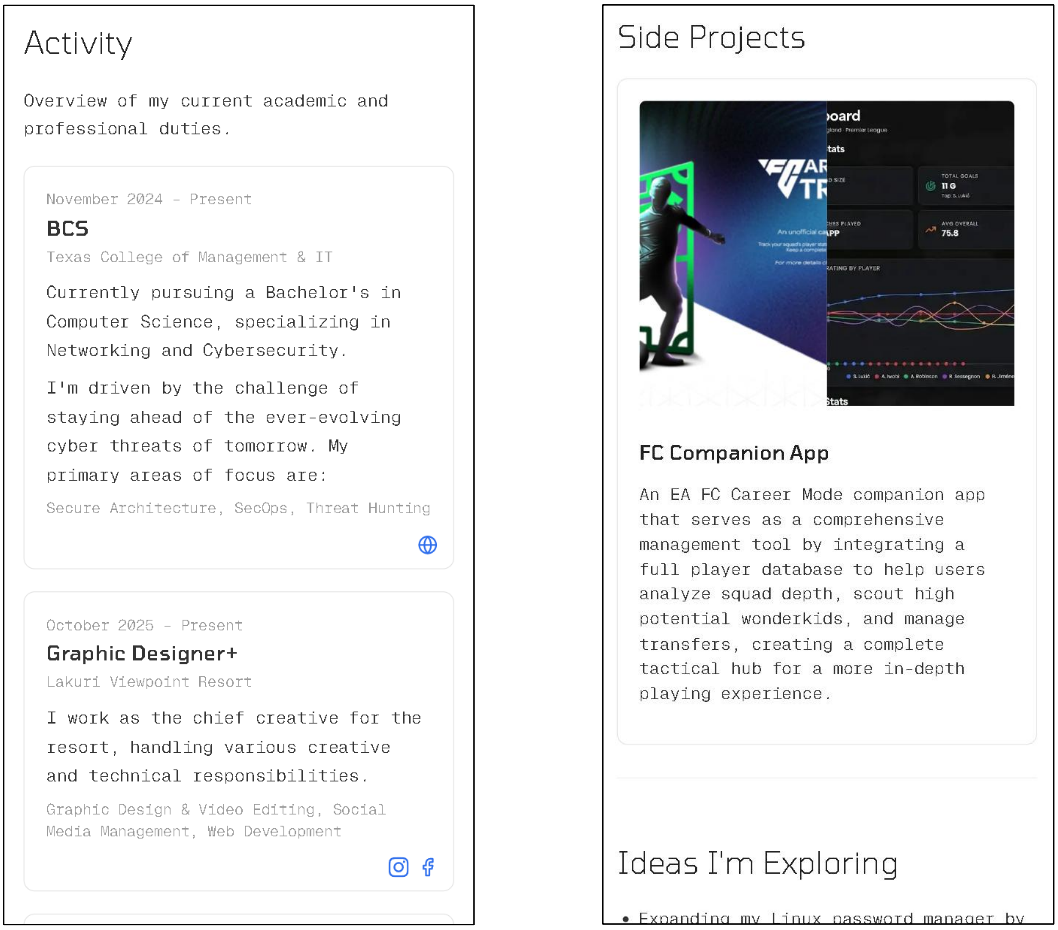

Figure 9. Mobile Views

Conclusion

Working on the Digital Identity Project meant taking everything I learned in class, semantic HTML, layout systems, typography, colour theory, and responsive design, and actually turning it into something real: a portfolio website that a recruiter or a collaborator could genuinely use to learn about my skills and my projects. Planning the page around the F-Layout pattern before I wrote any code meant that the most important parts of my story would land exactly where a visitor's eyes naturally go, and keeping my colours and fonts simple made the whole site feel professional and easy to follow rather than cluttered. Between the finished website, the GitHub repository behind it, and the live deployment, I feel like this project meets everything the assignment brief asked for.

Appendix

Website's Lighthouse Score

The completed portfolio website was evaluated using Google Lighthouse to assess its overall quality in terms of performance, accessibility, best practices, and search engine optimisation. The website achieved an overall *Performance score* of 97, *Accessibility score* of 91, *Best Practices* score of 100, and *SEO* score of 100, demonstrating that it is fast, follows modern web development standards, and is well optimised for search engines. Key performance metrics included a *First Contentful Paint (FCP)* of 1.1 seconds, *Largest Contentful Paint (LCP)* of 1.1 seconds, *Total Blocking Time (TBT)* of 30 ms, and a *Cumulative Layout Shift* (CLS) of 0.11, indicating a responsive and smooth user experience.

The report also highlighted a few areas for improvement, including increasing colour contrast for better accessibility, reducing unused JavaScript, optimising cache lifetimes, and improving image delivery. Overall, the Lighthouse results confirm that the website performs efficiently while adhering to recognised web accessibility, performance, and SEO best practices.

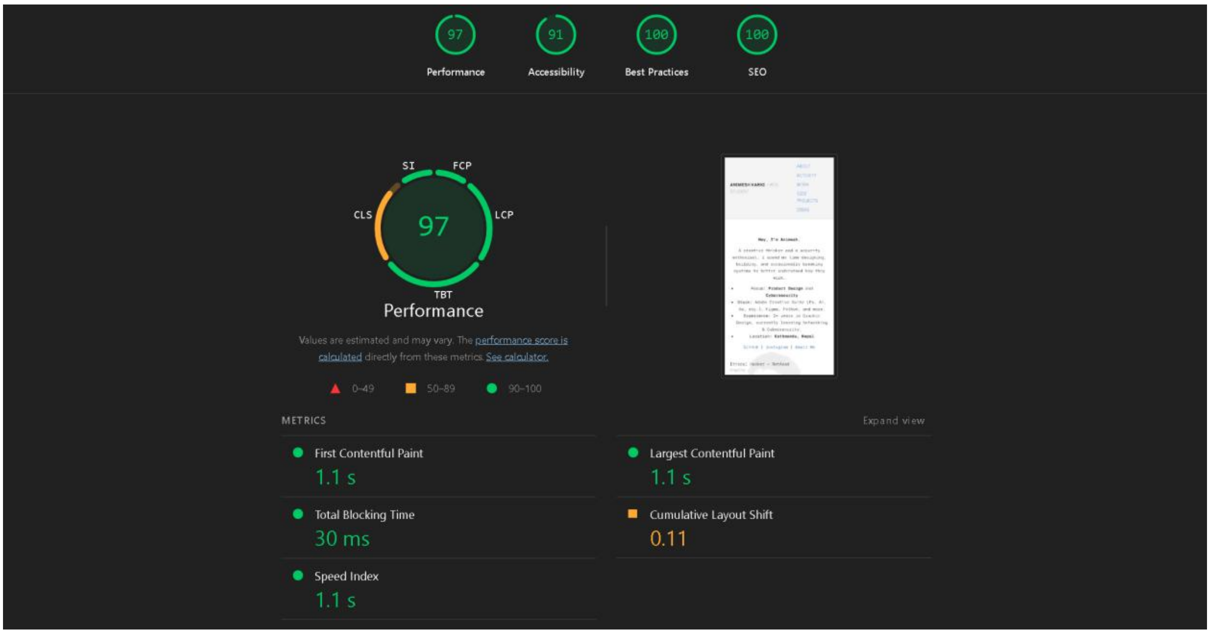

Figure 10.Webpage's Lighthouse Score